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Uncovering Market Inefficiencies in New York City Apartment Pricing: A Geospatial and Demographic Modeling Approach

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ABSTRACT

This paper examines how reported crime and neighborhood characteristics relate to mean gross rent across New York City's Neighborhood Tabulation Areas (NTAs). We spatially joined NYPD complaint records to NTA boundaries, converted reported incidents occurring in 2023 to rates per 1,000 residents, and combined those measures with demographic and housing estimates from the American Community Survey (ACS) via NYC Population FactFinder. The analysis includes 180 NTAs in the Bronx, Brooklyn, Manhattan, and Queens. A stepwise-selected linear regression model containing 11 demographic and housing variables had an in-sample R-squared of 0.9649; crime measures were not retained by that selection procedure. A separate model using only 2023 felony, misdemeanor, and violation rates had an R-squared of 0.328. We then mapped differences between observed and model-predicted mean gross rent to identify neighborhoods for further investigation. Several NTAs in Queens had observed rents below the model's predictions. These residuals indicate associations within the analyzed data, not verified discounts on currently available apartments or evidence of future price changes. The findings illustrate how neighborhood characteristics can inform rental-market analysis while also showing the limits of 2023 crime rates as standalone predictors in this dataset.

INTRODUCTION

This study examines associations between reported crime, socioeconomic and housing characteristics, and mean gross rent across New York City's Neighborhood Tabulation Areas (NTAs). We also identify neighborhoods whose observed mean gross rent differs substantially from a model prediction based on the included characteristics.

Housing costs shape where people can live, and differences among neighborhoods are relevant to renters, housing analysts, and policymakers. A comparison of observed and predicted rent may highlight areas worth investigating further, although a model residual alone cannot establish that an available apartment is a bargain or that a neighborhood is mispriced. The analysis may be useful to several audiences:

- Renters can identify neighborhoods to investigate further for affordability and other priorities.
- Real estate investors can examine neighborhoods with rents below the model's predictions as a starting point for additional market analysis.
- Housing analysts and policymakers can assess how reported crime relates to rent alongside demographic and housing measures.
- Property managers and developers can consider how neighborhood characteristics relate to observed rents, subject to the limits of an area-level model.

The study combines geographic data preparation with regression and residual mapping. We compare the explanatory power of crime measures alone with that of a model selected from demographic and housing variables, then map the differences between observed and predicted rents. This approach identifies geographic patterns for further investigation; it does not establish why individual neighborhoods have a positive or negative residual.

DATA AND METHODS

This study uses NYPD complaint records from NYC Open Data and American Community Survey (ACS) five-year estimates covering 2020-2024, accessed through NYC Population FactFinder. Although the prepared monthly crime dataset spans 2010-2024, the analysis uses reported incidents occurring in 2023. We selected 2023 because it provides a recent, complete calendar year for consistent comparisons

across NTAs and falls within the ACS estimation period. It also follows the initial COVID-19 disruption of 2020-2021, although pandemic-related effects may have persisted. The complaint records include incident locations and offense categories, allowing us to aggregate felony, misdemeanor, and violation counts by NTA. We excluded Staten Island from the analysis because its transit access and neighborhood characteristics differ from those of the other four boroughs.

The ACS-based dataset supplies the outcome, Mean Gross Rent, and demographic and housing measures at the NTA level. Mean gross rent describes the rent and estimated utilities paid by occupants of renter-occupied homes; it is not a measure of current asking rents for available apartments. The candidate variables include measures of age, income, employment, housing age, and housing occupancy.

We used an ArcGIS spatial join to associate incident coordinates with NTA boundaries, then aggregated the joined records occurring in 2023 by NTA and offense category (felony, misdemeanor, or violation). The resulting crime measures were joined to the ACS-based data using NTA names, producing 180 NTAs in the analysis dataset.

We expressed 2023 reported incidents per 1,000 residents by dividing each NTA's count for each offense category by its ACS population estimate and multiplying by 1,000. These measures represent a single calendar year, rather than cumulative counts or averages across 2010-2024. Selected demographic and housing totals were converted to percentages; for example, units built in 1939 or earlier were expressed as a percentage of housing units. The crime measures describe reported activity within an NTA and do not measure each resident's individual exposure to crime.

STATISTICAL ANALYSIS AND RESULTS

The dataset contained more than 70 candidate predictors and 180 observations. We used stepwise selection to choose variables for a least squares regression of mean gross rent on demographic, housing, and crime measures. We also fitted a separate least squares model using the three population-adjusted crime categories. Because selection and fit were assessed on the same observations, the reported fit statistics describe this sample and have not been validated on new data.

The selected model had an in-sample R-squared of 0.9649 and an adjusted R-squared of 0.9622. Its 11 selected demographic and housing variables accounted for approximately 96% of the variation in mean gross rent among these 180 NTAs. This fit does not by itself show that the variables cause rent differences or that the model will predict new observations equally well.

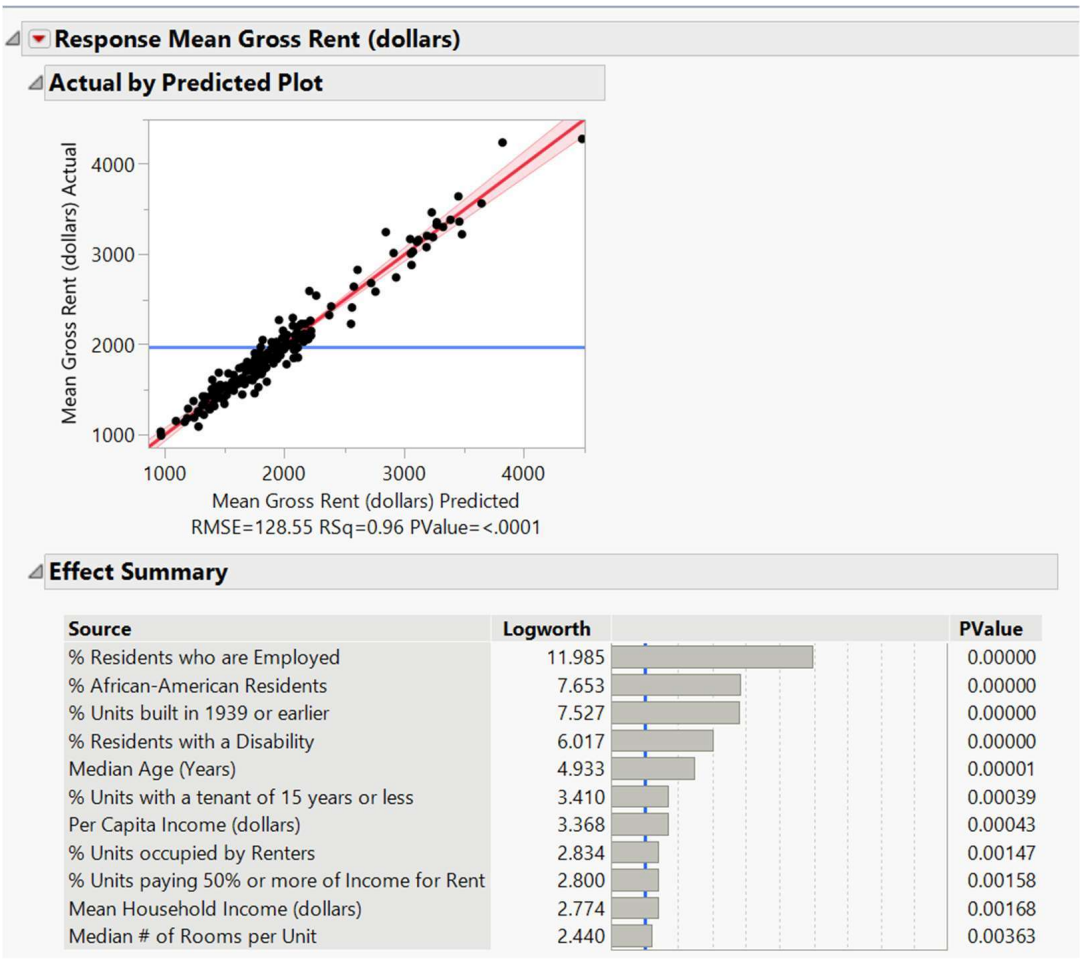
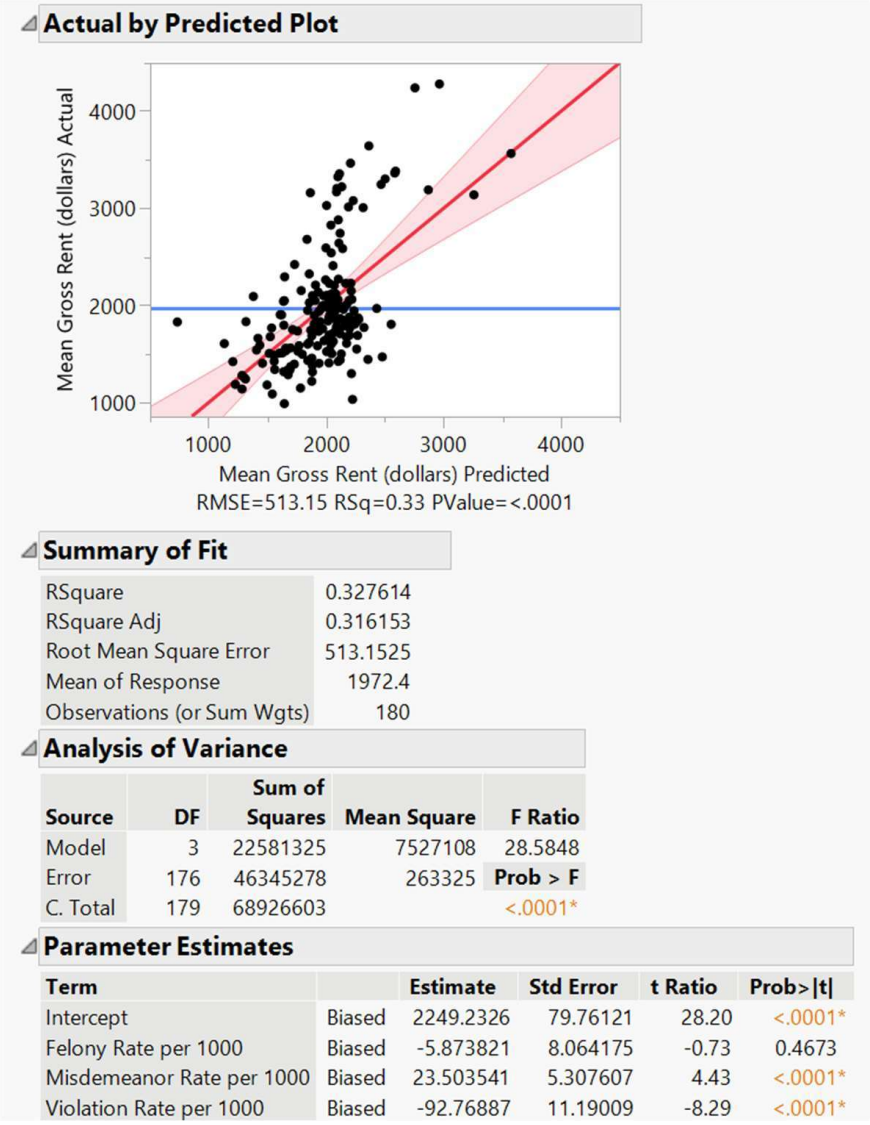


Figure 1. Actual Versus Predicted Rent and Selected Model Effects

No 2023 crime rate was retained in the final stepwise-selected model. Under this selection procedure, the three crime categories did not add enough explanatory value alongside the selected demographic and housing measures to remain in the model. Figure 1 ranks the selected effects by statistical significance; it does not show the direction of their estimated associations. Employment, income, housing age, and occupancy-related measures appear among the selected variables. Measures concerning race and disability also appear, reflecting area-level associations that should not be interpreted as causal effects of residents' identities. Housing history, location, and other unmeasured factors may contribute to those patterns.

To examine the crime measures separately, we fitted a regression of Mean Gross Rent on reported felony, misdemeanor, and violation incidents occurring in 2023 per 1,000 residents. The crime-only model was statistically significant overall ($p < 0.0001$) but had an R-squared of 0.328. Thus, the three reported-crime categories together accounted for about one-third of the variation in mean gross rent in this sample, substantially less than the selected demographic and housing model.

In the crime-only model, the felony-measure coefficient was not statistically distinguishable from zero after accounting for the other two crime categories ($p = 0.4673$). The misdemeanor and violation coefficients were statistically significant but had opposite signs in this specification. This pattern cautions against treating their sum as a straightforward measure of neighborhood safety or interpreting these associations as causal effects. Other neighborhood characteristics, including geography and housing supply, may also be related to rent.



Summary of Fit

RSquare	0.327614
RSquare Adj	0.316153
Root Mean Square Error	513.1525
Mean of Response	1972.4
Observations (or Sum Wgts)	180

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	22581325	7527108	28.5848
Error	176	46345278	263325	Prob > F
C. Total	179	68926603		<.0001*

Parameter Estimates

Term		Estimate	Std Error	t Ratio	Prob>|t|
Intercept	Biased	2249.2326	79.76121	28.20	<.0001*
Felony Rate per 1000	Biased	-5.873821	8.064175	-0.73	0.4673
Misdemeanor Rate per 1000	Biased	23.503541	5.307607	4.43	<.0001*
Violation Rate per 1000	Biased	-92.76887	11.19009	-8.29	<.0001*

Figure 2. Crime Only Model Fit and Parameter Estimates
Crime rates represent reported incidents occurring in 2023 per 1,000 residents.

We calculated each NTA's residual as observed mean gross rent minus the prediction from the selected model. A negative residual means the observed NTA average is below the model's prediction. We mapped and ranked these residuals to examine geographic patterns that the included predictors did not capture. Residuals may reflect omitted variables, measurement differences, or model specification; they do not demonstrate that current listings are underpriced.

The largest negative residual in Figure 3 belongs to Astoria (Central), at approximately $323 per month below the model prediction for mean gross rent. Pomonok-Electchester-Hillcrest, Cambria Heights, and Woodside are also among the NTAs with large negative residuals. These differences identify places for closer examination, rather than verified discounts available to an apartment hunter. The figure also includes NTAs outside Queens.

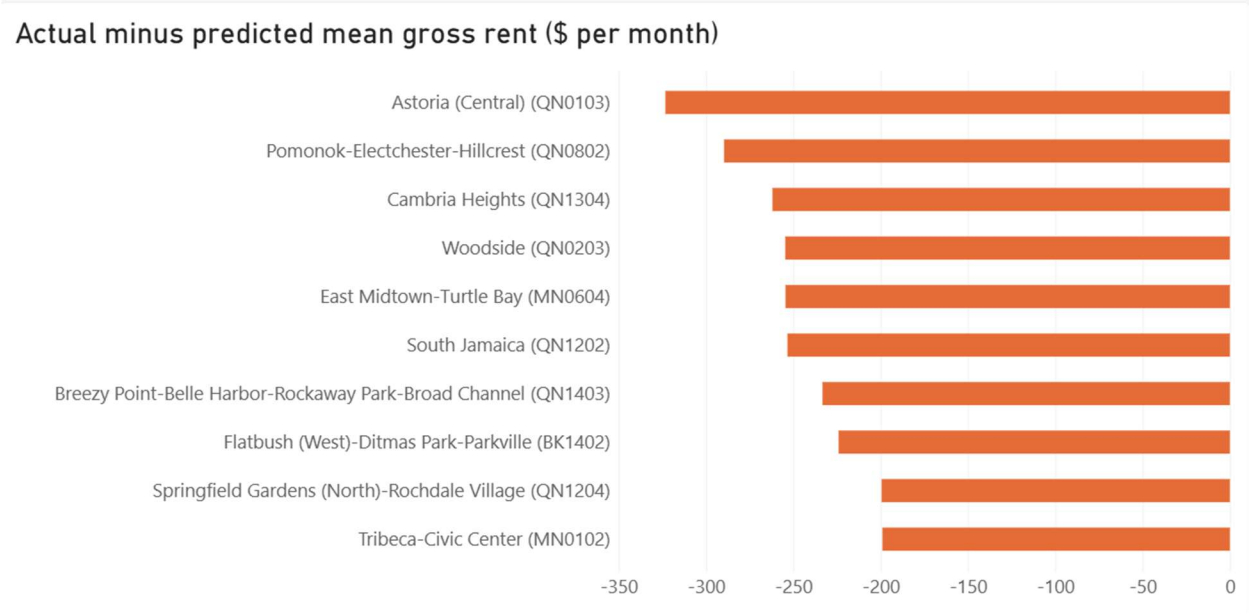


Figure 3. Neighborhoods with the Largest Negative Model Residuals

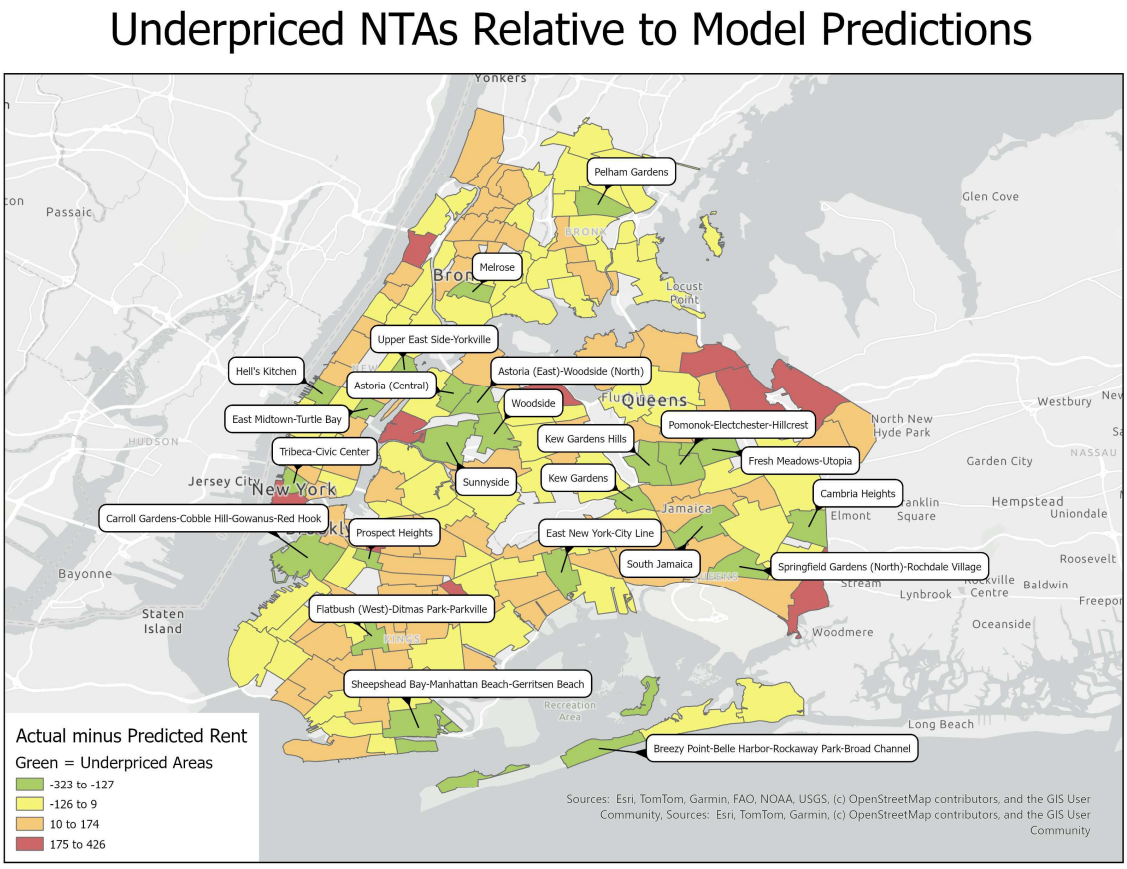


Figure 4. Geographic Distribution of Rent Residuals

Figure 4 maps each NTA's rent residual, calculated as observed minus model-predicted mean gross rent. Green areas have the largest negative residuals, indicating rents below the model's predictions; red areas have the largest positive residuals. Several negative residuals appear in Queens, with others in Brooklyn

and the Bronx. These geographic patterns may reflect factors the model does not fully capture, such as transit access, location, or housing stock. The map does not establish that currently available apartments in those areas are underpriced.



Figure 5. Neighborhoods with Lower Reported Crime and Below-Predicted Rent
Horizontal axis: reported misdemeanors and violations occurring in 2023 per 1,000 residents.

Figure 5 shows the lower-left portion of the full scatter plot, highlighting selected NTAs with relatively low reported misdemeanor and violation incidents per 1,000 residents in 2023 and mean gross rents below model predictions. Points farther left have fewer reported incidents relative to population for these two offense categories, while points lower on the chart have more negative rent residuals, indicating larger differences below predicted rent. For example, Astoria (Central) has a residual of approximately -$323 per month and about 33 reported misdemeanors and violations per 1,000 residents; Flatbush (West)-Ditmas Park-Parkville has a residual near -$224 per month and about 26 incidents per 1,000 residents. The horizontal axis excludes felonies. This illustrative subset highlights neighborhoods with lower reported crime measures and below-predicted rent; it does not establish that these neighborhoods have the lowest actual rents, that available apartments are affordable, or that a neighborhood is safe for an individual renter.

CONCLUSION

This study examined how reported crime, demographic characteristics, and housing measures relate to mean gross rent across 180 New York City NTAs. 2023 crime rates alone had limited explanatory power in this sample, while a model selected from demographic and housing measures fit the observed NTA averages closely. These are descriptive associations rather than estimates of causal effects.

The crime-only model had an R-squared of 0.328, and the felony-measure coefficient was not statistically significant after accounting for the 2023 misdemeanor and violation rates. The stepwise-selected model had an in-sample R-squared of 0.9649. The difference in fit suggests that rent varies with characteristics beyond the three reported-crime categories. It does not establish that crime has no effect on housing decisions, nor does it isolate the effects of location, transit, housing supply, or income. The selected model's high fit should also be interpreted cautiously because variable selection and evaluation used the same 180 observations.

Several Queens NTAs, including Astoria (Central), Cambria Heights, and Woodside, had observed mean gross rents more than $250 below the model's predictions. Renters or analysts could investigate current

listings and additional neighborhood information in these areas. Investors would likewise need property-level prices, operating costs, and market evidence before drawing investment conclusions. A negative NTA residual is not a promised monthly saving or evidence that a future market correction will occur.

The crime measures describe reported incidents occurring in 2023, whereas the ACS estimates summarize 2020-2024. Although 2023 falls within that estimation period, the measures do not represent identical time windows. A single year of crime data may reflect unusual conditions and may not represent longer-term patterns. The population denominator is an ACS five-year estimate rather than a population count specific to 2023. In addition, the percentage of units paying at least 50% of income toward rent is related to the rent outcome by construction and may contribute to the model's high in-sample fit. A model excluding that predictor has not been evaluated in this study.

The analysis is cross-sectional and evaluates model fit on the same data used for variable selection. Future work could test predictive performance on held-out NTAs or another time period and examine spatial dependence among neighboring areas. It could also add transit-access measures, such as walking distance to stations, and compare ACS rent estimates with contemporaneous asking rents. These steps would help determine whether the observed residual patterns persist and how much of them can be explained by omitted geographic factors.

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